# EasySeg: An Error-Aware Domain Adaptation Framework for Remote Sensing Imagery Semantic Segmentation via Interactive Learning and Active Learning

Liangzhe Yang, Hao Chen, Anran Yang, and Jun Li

***Abstract*— Semantic segmentation of remote sensing images has attracted much attention for its wide applications. While deep learning models have shown impressive performance in this task, challenges arise when applying them to data from other domains without fine-tuning, due to domain gaps. Domain adaptation (DA) has emerged as a solution to bridge this gap. Existing works mainly focus on unsupervised DA (UDA), which lags far behind fully supervised models. However, active DA (ADA) methods focused on natural images face challenges when applied to remote sensing images due to pronounced domain gaps and error unawareness problems. In this work, we propose a novel error-aware DA framework for remote sensing imagery semantic segmentation, called EasySeg, via interactive learning and active learning. First, we introduce a point-level labeling strategy, named "See-First-Ask-Later" (SFAL), combining both interactive and active learning manners, allowing obvious errors and information-rich pixels to be annotated easily and efficiently. Then, we introduce an interactive semantic segmentation network (ISS-Net), which can perform automatic semantic segmentation and interactive refinement. Based on the acquired target point-level labels, ISS-Net generates dense and accurate pseudo-labels to enhance DA performance through retraining with consistency regularization. Comprehensive experiments on two tasks demonstrate that our method outperforms the state-of-the-art ADA methods in terms of overall accuracy (OA), mean accuracy (MA), $F1$ score, and mean intersection of union (mIoU) with lower labeling costs, even surpassing some fully supervised models. The source code of EasySeg is freely available at https://github.com/Yangliangzhe/EasySeg.**

***Index Terms*— Active learning, domain adaptation (DA), interactive learning, remote sensing images, semantic segmentation.**

## I. Introduction

SEMANTIC segmentation of remote sensing images is a fundamental research area with significant applications in urban planning [1], land-cover mapping [2], and land use analysis [3]. Recent years have witnessed a proliferation of deep learning-based works for remote sensing imagery semantic segmentation, achieving remarkable results on various benchmarks. However, the difference in remote sensing sensors, monitoring objectives, and observation altitudes, makes the characteristics such as spatial resolution, spectral channels, texture, and scene context vary. These diverse variations lead to substantial differences in the feature distributions of various remote sensing imagery datasets. The difference in feature distribution between the source and target data is referred to as the domain gap between the two datasets [4], which presents a notable challenge for the semantic segmentation of remote sensing images due to their sensitivity to changes in the data domain [5]. Specifically, when the semantic segmentation methods [7], [8] encounter target data with a substantial domain gap from their training data, they experience a pronounced deterioration in performance, thereby posing a significant hindrance to the practical applications of remote sensing imagery semantic segmentation.

Since acquiring pixel-wise annotations for remote sensing images is costly and time-consuming, domain adaptation (DA) has emerged as one of the promising solutions to bridge the domain gap. DA in semantic segmentation addresses the issue by transferring knowledge from label-rich data (source data) to label-scarce data (target data) [6], [9]. The mainstream DA semantic segmentation for remote sensing images primarily focuses on unsupervised DA (UDA) [5], [10], [11], [12], leveraging labeled source data and unlabeled target data. According to their adaptation strategies, UDA can generally be divided into adversarial learning [13], [14] and self-training [15], [16]. The former employs adversarial networks to narrow the feature distribution gap between source and target domains, while the latter involves model retraining through generated pseudo-labels. Despite the potential of UDA to mitigate the need for extensive manual annotations, its performance still lags far behind fully supervised methods [9], [17], [18]. In practical contexts, annotations are not entirely out of reach, and the focus instead lies in maximizing the segmentation performance in the target domain while minimizing manual annotation costs. Recently, some efforts have attempted to incorporate a few labels into the target domain for DA semantic segmentation. Wang et al. [19] proposed a semi-supervised DA method,

Manuscript received 22 October 2023; revised 9 February 2024 and 7 April 2024; accepted 30 April 2024. Date of publication 9 May 2024; date of current version 21 May 2024. This work was supported by the National NSF of China under Grant U19A2058, Grant 41971362, Grant 41871248, and Grant 62106276. *(Corresponding author: Hao Chen.)*

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Digital Object Identifier 10.1109/TGRS.2024.3399260

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which involves image-level samples randomly selected from the target domain. Paul et al. [20] randomly sampled pixels from each class in the target data for annotation, using these sparse point-level annotations to enhance performance. Considering the inefficiency of these random labeling methods, some works [9], [18] have employed active learning to obtain information-rich labels in the target data.

Active DA (ADA) aims to uncover the most informative unlabeled data to maximize model performance while minimizing manual labeling costs. Most ADA methods use model uncertainty [9], [21] or data diversity [22], [23] as indicators to select informative data for labeling. As to semantic segmentation, Guan and Yuan [9] selected target images for annotation based on the mean entropy of model predictions. Wu et al. [23] annotated regions segmented by superpixels in the target data based on domain density. Since acquiring labels for dense prediction tasks such as semantic segmentation is more resource-intensive and time-consuming than image classification, several recent ADA works [5], [18] have opted to label specific pixels (point-level), which is easier than labeling entire images (image-level) or some regions (region-level). In this work, we focus on the point-level ADA for remote sensing imagery semantic segmentation.

While several methods have been explored in ADA semantic segmentation for natural images, challenges persist in their applications to remote sensing images. We observe that these challenges likely arise from the pronounced domain gap in remote sensing images, including differences in remote sensing sensors, spatial resolution, imaging spectrum, and imaging angle, which are different from those in natural images. Coupled with limited annotations, the model frequently misunderstands objects in the target data, causing conspicuous errors with high model confidence. Meanwhile, existing active learning frameworks tend to favor uncertain predictions for selection, making it challenging to identify these obvious yet model-confident errors. This error unawareness issue hinders the timely rectification of model misunderstandings, introducing biases in active learning and consequently limiting the improvement in model performance with the acquired annotations. Moreover, unlike methods acquiring image-level or region-level labels which can directly provide contextual information, point-level ADA semantic segmentation methods encounter more significant challenges in rectifying these biases due to the sparse nature of point-level labels.

To address the error unawareness issue, we consider interactive learning. In interactive learning, pixels at the center of the largest mislabeled regions are sequentially labeled. This corrective labeling process passively guides the model to learn the target domain. However, when predictions are accurate enough, the mislabeled regions become small and challenging to identify, which increases the annotation burden. Moreover, these passively discovered small errors may have lower value than the pixels selected through active learning. Based on this observation, we propose an error-aware DA framework named EasySeg for remote sensing imagery semantic segmentation, incorporating both the advantages of interactive learning and active learning. First, we propose a novel point-level labeling strategy called "See-First-Ask-Later" (SFAL). In this strategy, users initially identify and label obvious mispredictions by simply clicking at the centers of the erroneous regions. Then, an active learning strategy is employed to query those information-rich pixels in predictions. The core of the strategy is to combine passive and active manners to make the acquired labels both error-aware and information-rich at a low cost. Second, considering the sparse nature of point-level labels in the target domain, we introduce a novel Interactive Semantic Segmentation Network for remote sensing imagery, called ISS-Net. Besides the ability of automatic semantic segmentation, ISS-Net can also accept acquired target labels to generate dense and precise pseudo-labels in an interactive semantic segmentation manner. On one hand, we retrain our ISS-Net through the pseudo-labels and consistency regularization [24] to enhance automatic semantic segmentation performance on the target domain. On the other hand, we design a click loss for ISS-Net to ensure the fusion of automatic semantic segmentation and interactive segmentation, preventing the network from disregarding interactive inputs.

Our main contributions can be summarized as follows.

1) We propose a novel DA framework named EasySeg for remote sensing imagery semantic segmentation using active learning and interactive learning, which allows obvious errors and information-rich pixels to be labeled efficiently at a low cost. To the best of our knowledge, it is the first work to combine active learning and interactive learning to do DA for remote sensing imagery semantic segmentation.
2) An effective point-level labeling strategy and a novel ISS-Net are proposed to mitigate the data gap from the source domain to the target domain.
3) Comprehensive experiments demonstrate the effectiveness and superiority of our EasySeg. Our approach outperforms other state-of-the-art methods with a lower labeling cost, even surpassing some fully supervised models.

The rest of this article is organized as follows. Section II presents the related work, Section III provides a detailed explanation of the proposed framework, Section IV presents the experimental details and results, Section V presents a discussion and finally, Section VI offers our conclusions and outlines future work.

## II. Related Work

### A. Domain Adaptation

DA can transfer knowledge from a label-rich domain to a label-scarce domain, which has been studied in many vision tasks such as classification, object detection, and semantic segmentation. The main challenge in DA is minimizing the domain gap between the source and target data. Recent UAD methods in semantic segmentation primarily fall into two categories: adversarial learning and self-training. Adversarial learning typically reduces the discrepancy through alignment at the input level [25], [26], intermediate feature level [13], [27], or output level [28], [29]. Self-training [30], [31] offers a competitive alternative, which retrains the model using

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pseudo-labels generated from confident predictions on the target data. As to UDA semantic segmentation for remote sensing images, Chen et al. [5] ensured semantic consistency through both global and category-level alignment, while Ma et al. [12] employed multidiscriminator techniques to align features extracted by transformers at both low-level and high-level. Zhang et al. [10] designed adversarial-based semantic- and entropy-level domain classifiers to reduce the locally semantic and globally structural feature differences of remote sensing images between domains. Zhu et al. [11] proposed an invariant feature memory module to mitigate the unfavorable impact of only representing the variant features from the currently limited inputs using an adversarial learning scheme.

Since UAD semantic segmentation lags far behind supervised learning [17], recently, some efforts [18], [19], [20] have attempted to incorporate a few labels into the target domain. Among them, ADA semantic segmentation [9], [18], which uncovers the most informative unlabeled data from the target domain to maximize model performance while minimizing manual labeling costs, has gained momentum for its efficacy in enhancing DA. In this work, we adopt a novel labeling strategy for ADA semantic segmentation of remote sensing images.

### B. Active Learning

The core of active learning is designing an effective labeling strategy to acquire the most informative annotations with minimal cost. Labeling strategies can commonly be categorized into two main approaches: uncertainty-based [9], [17] and density-based [22], [23] methods. In the ADA, previous works mainly focus on classification tasks [21], [32]. Ning et al. [55] and Shin et al. are the first to study ADA for semantic segmentation [18]. Ning et al. used clustering-based anchors to select representative target images that are the most complimentary to the source domain. However, such an image-level method is highly inefficient, since an extensive labeling budget is wasted on the redundant image areas [18]. Shin et al. [17] and Xie et al. [18] proposed two efficient labeling options, point-level and region-level. Shin et al. used the inconsistency in bi-classifier predictions as the indicator to select labeling objects, while Xie et al. selected the pixels based on region impurity and prediction uncertainty. Guan and Yuan employed an image-level labeling strategy based on the mean entropy of prediction and combined two robust semi-supervised methods [33], [34] to improve performance. Wu et al. selected representative target regions generated by superpixel segmentation based on domain density.

Existing active learning methods commonly overlook significant yet confident errors in model predictions, leading to substantial noise when applied to remote sensing images due to the pronounced domain gap. Compared to image-level and region-level methods, this noise is disastrous for point-level methods due to the larger biases brought about by sparse labeling. In this work, we tailor our approach to point-level annotation scenarios and introduce an error-aware labeling strategy called SFAL. It promptly corrects the model's misconceptions of the target data, thus enhancing ADA performance.

### C. Interactive Learning

Interactive segmentation can output class-agnostic masks in a desired manner based on user interactions, enabling the refinement of existing masks through multiple human–computer interactions [35]. Many efforts [36], [37] have been invested in improving the labeling efficiency in interactive segmentation. To make point-level annotations dense, we believe that generating pseudo-labels through interactive segmentation is a promising approach. However, interactive segmentation models are semantically agnostic and face domain gaps as well. To address these challenges, we merge automatic semantic segmentation with interactive segmentation, proposing a unified ISS-Net. ISS-Net encompasses the capabilities of semantically automatic prediction and interactive refinement, while simultaneously achieving DA for both tasks.

## III. Method

In this section, we provide a mathematical description of our problem and outline our proposed framework in Section III-A. Subsequently, in Sections III-B–III-E, we will present the four key modules of our framework. In Section III-F, we detail the training process of our framework.

### A. Overview

We focus on the problem of ADA semantic segmentation of remote sensing images. In ADA semantic segmentation, we have labeled source data $S = (X^s, Y^s)$ and unlabeled target data $T = (X^t, Y^t_{\text{point}})$, where $Y^s$ represents the pixel-wise labels of the $C$ defined classes and $Y^t_{\text{point}}$ represents the point-level labels that are initialized as $\varnothing$ and provided by human. It's worth noting that in ADA semantic segmentation, achieving alignment between the defined classes of the source and target domains is necessary. The goal is to learn a model $G_\theta: X \rightarrow Y$ with the parameter $\theta$ that performs well on target domain segmentation, with limited annotations.

The overview of the proposed EasySeg is illustrated in Fig. 1. It consists of Fig. 1(a) DA semantic segmentation learning, Fig. 1(b) image labeling via SFAL, Fig. 1(c) interactive semantic segmentation learning, and Fig. 1(d) pseudo-labels retraining with consistency regularization. Before labeling the target data, the model undergoes warm-up training in the source domain, utilizing Fig. 1(a) and (c). Subsequently, Fig. 1(c) is used for incremental labeling of the target data in specific rounds. After each labeling round, DA training is conducted using Fig. 1(a), (c), and (d) to transfer knowledge to the target domain. Our ISS-Net incorporates two capabilities: automatic semantic segmentation and interactive refinement. The former receives images and directly generates predictions, while the latter refines predictions based on optional interactive inputs. In Fig. 1(a), ISS-Net performs automatic semantic segmentation. Fig. 1(c) focuses on training ISS-Net to possess interactive refinement ability using source data $(X^s, Y^s)$ and interactive inputs. In Fig. 1(d), we leverage the interactive refinement ability of ISS-Net to enhance the model's performance in the target domain based on the labeled target data

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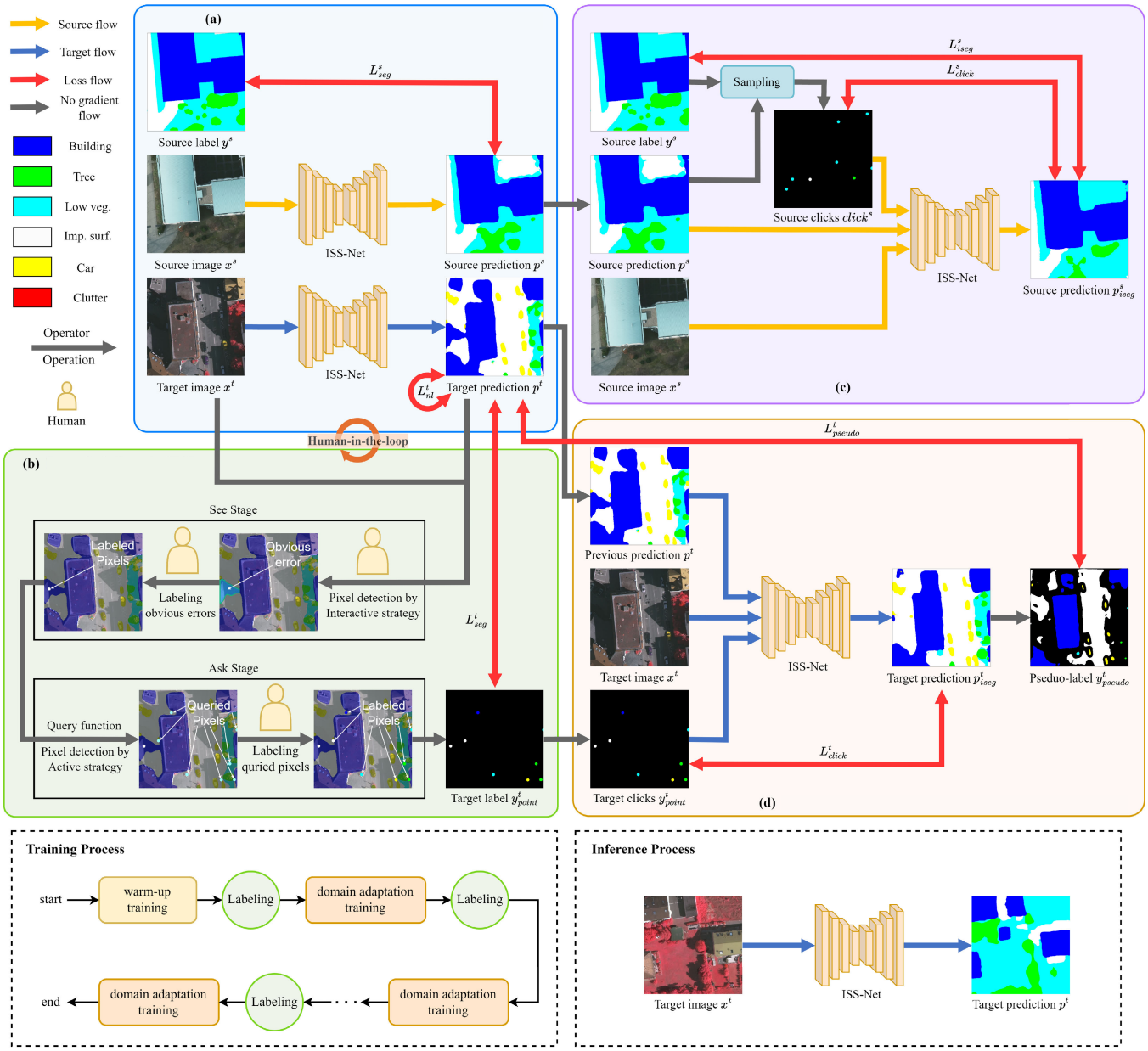



Fig. 1. Overview of the our EasySeg. All ISS-Nets share weights. Optional inputs are automatically initialized to zero matrices by ISS-Net when they are empty (labels are zoomed-in view eight times for better visualization. The consistency regularization is not shown above for clarity, and it can be found in Fig. 3). (a) DA semantic segmentation learning. (b) Image labeling via SFAL. (c) Interactive semantic segmentation learning. (d) Pseudo-labels retraining with consistency regularization.

$(X^t, Y^t_{\text{point}})$. During the inference process, we produce the target predictions based on the automatic semantic segmentation ability of ISS-Net.

### B. DA Semantic Segmentation Learning

In the warm-up training phase, Fig. 1(a) trains the model to acquire knowledge about semantic segmentation of remote sensing images from the image-label pairs in the source domain. After the first labeling round, Fig. 1(a) promptly includes the target data in supervised training, i.e., the DA stage. Specifically, we sample data of batch size from both the source and target data. Then, the model is updated based on the loss generated from these samples. Unlike the densely labeled source labels, the ground truth of the target data consists of only a few sparse pixels, with others set as $\text{ignore}_{\text{label}} = 255$. Throughout these two stages, we utilize a cross-entropy loss with ignore_label $= 255$ to train our model on both source and target data. Hence, the semantic segmentation losses, $L^s_{\text{seg}}$ for the source data and $L^t_{\text{seg}}$ for the target data, can be calculated as follows:

$$\rho_c\left(y_{i,j}\right) = \begin{cases} 1, & y_{i,j} = c \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

$$L_{\text{seg}} = \frac{1}{\sum_{i,j,c}^{H,W,C} \rho_c\left(y_{i,j}\right)} \sum_{i,j}^{H,W} \sum_{c}^{C} \rho_c\left(y_{i,j}\right) \log\left(p_{i,j,c}\right) \tag{2}$$

where $y_{i,j}$ represents the pixel at position $(i, j)$ in the ground truth $y \in \mathbb{R}^{H,W}$, and $p_{i,j,c}$ $(p \in \mathbb{R}^{H,W,C})$ represents the

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probability of class $c$ at position $(i, j)$ in the prediction. That is, in the warm-up training phase when there are no target labels, the value of the loss $L^t_{\text{seg}}$ remains constant at 0. When entering the DA training phase, the loss $L^t_{\text{seg}}$ is added to supervise the model training.

In the early stage of training, the low probabilities of pixels in the prediction indicate specific absent classes [18], [38], referred to as negative pseudo labels. For instance, though it is challenging to identify the class of a pixel with prediction values as [0.48, 0.49, 0.03], we can clearly know that it does not belong to the class with a score of 0.03. To apply this insight to target data, we add the negative learning loss into the above two training stages. First, we obtain the negative pseudo labels as follows:

$$\pi\left(p^t_{i,j,c}\right) = \begin{cases} 1, & p^t_{i,j,c} < \tau_{\text{nl}} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $\tau_{\text{nl}}$ represents the negative threshold, and we follow [18] to set $\tau_{\text{nl}} = 0.05$. Then, we introduce the target negative learning loss $L^t_{\text{nl}}$ into DA semantic segmentation learning, as shown in Fig. 1(a). It is calculated as follows:

$$L^t_{\text{nl}} = \frac{1}{H \cdot W} \sum_{i,j}^{H,W} \sum_{c}^{C} \pi\left(p^t_{i,j,c}\right) \log\left(1 - p^t_{i,j,c}\right) \tag{4}$$

where $p^t_{i,j,c}$ $(p \in \mathbb{R}^{H,W,C})$ represents the probability of class $c$ at position $(i, j)$ in the target prediction. The weight of target negative learning loss $L^t_{\text{nl}}$ is denoted as $\lambda^t_{\text{nl}}$ and is set to 1.0 following [18].

### C. Image Labeling via SFAL

Due to the significant domain gap in remote sensing images, the model's understanding of target data is rather limited, resulting in high-confidence but large-region errors. Unfortunately, these errors, although they are obvious and fundamental, are difficult to select by existing active learning methods, leading to a significant bias in the trained model. To make the model error-aware for target data, we introduce a point-level labeling strategy called SFAL, as shown in Fig. 1(b). It consists of two stages: the "See" stage and the "Ask" stage. This strategy aims to address the above challenge and improve the model's understanding of the target data, thereby promoting enhanced performance and reliability.

Image labeling is performed only at specific training iterations, and the total number of labeled pixels, denoted as $N_p$, is typically restricted. For a fair comparison, we follow [18] to label 40 pixels per image over five rounds, with 8 pixels annotated in each round.

*1) See Stage:* The "See" stage focuses on identifying fundamental errors made by the model. During the image labeling process, human annotators first evaluate the quality of the predicted masks overlaid on the images. When obvious and absurd errors are detected, annotators place a specific label at the center of the large-scale mislabeled region by simply clicking the mouse. To simulate this process computationally, we introduce a function $\text{score}_i(M)$ to assess the significance of errors in class $i$. It is calculated as follows:

$$\delta_{i,j}(M) = \max\left(\frac{M_{i,j}}{\sum_j^C M_{i,j}}, \frac{M_{i,j}}{\sum_i^C M_{i,j}}\right) \tag{5}$$

$$\text{score}_i(M) = \max\left(\delta_{i,1}(M), \delta_{i,2}(M), \ldots, \delta_{i,C}(M)\right) \tag{6}$$

where $M$ is the confusion matrix and $M_{i,j}$ represents the number of pixels incorrectly classified as class $j$ when they actually belong to class $i$.

When the $i \to j$ mislabeled regions are significant compared to the ground truth mask of class $i$ or class $j$, the error rate, denoted as $\delta_{i,j}$, will be large. We define the error score $\text{score}_i$ for class $i$ as the maximum value of all error rates associated with class $i$. If the error score for a certain class exceeds the predefined threshold $\epsilon_{\text{error}}$, the computer will automatically sample a pixel near the center of the mislabeled region for labeling. Therefore, in this stage, we will sample $n_{\text{see}}$ $(0 \le n_{\text{see}} \le C)$ pixels.

*2) Ask Stage:* Different from the "See" stage, the "Ask" stage focuses more on the challenging pixels commonly located at object boundaries [18]. Since these pixels are difficult for humans to identify, we select them using active learning. We adopt the point-level query function based on region impurity and prediction uncertainty proposed in the article [18]. The region impurity of each pixel is calculated as follows:

$$r^t_{i,j} = -\sum_{c=1}^{C} \frac{\left|N^c_k(i,j)\right|}{\left|N_k(i,j)\right|} \log\left(\frac{\left|N^c_k(i,j)\right|}{\left|N_k(i,j)\right|}\right) \tag{7}$$

where $N_k(i, j) = \{|u - i| \le k, |v - j| \le k\}$ represents the square region of the target image $x^t$ with side length $k$ centered at position $(i, j)$. $N^c_k(i, j)$ is the subregion of class c in the region $N_k(i, j)$, and $|\cdot|$ denotes the number of pixels in the region.

The prediction uncertainty is calculated based on the entropy of each pixel as follows:

$$u^t_{i,j} = -\sum_{c=1}^{C} p^t_{i,j,c} \log\left(p^t_{i,j,c}\right). \tag{8}$$

Then, the query value of pixel $(i, j)$ is calculated as follows:

$$\text{query}_{i,j} = u^t_{i,j} \cdot r^t_{i,j}. \tag{9}$$

In each labeling round, we select the top $n_{\text{ask}} = 8 - n_{\text{see}}$ pixels with the highest query values.

The process of the "See" stage and the "Ask" stage is shown in Fig. 1(b). In the SFAL labeling strategy, we first perform the "See" stage to annotate the obvious errors and correct the learning bias, and then perform the "Ask" stage to mine the information-rich pixels for annotation through the query function.

### D. Interactive Semantic Segmentation Learning

To generate dense and accurate pseudo-labels, we fuse automatic semantic segmentation with interactive segmentation to construct an ISS-Net, enabling refinements through human interventions in the automatic segmentation results while preserving semantic information in the interactive segmentation.

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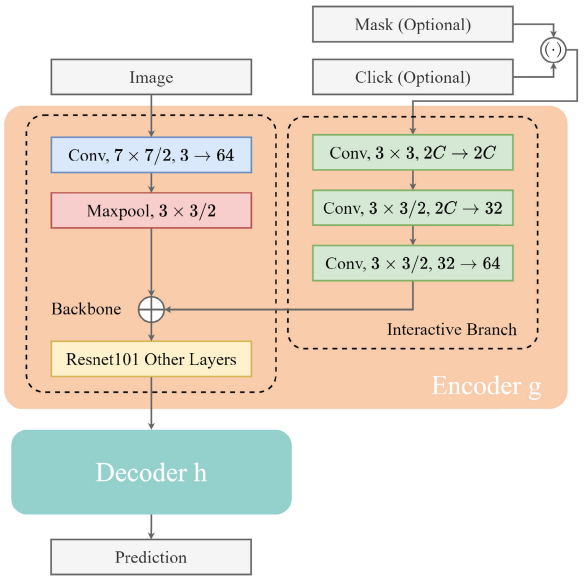



Fig. 2. Illustration of ISS-Net with resnet101 [45] as backbone. In this implementation, we use deeplabv2 [46] or deeplabv3plus [47] as the decoder. The symbols "+" and "(·)" indicate addition and concatenation operations, respectively. "Conv, $7 \times 7/2$, $3 \rightarrow 64$" means $7 \times 7$ conv layer with stride $= 2$, $\text{channel}_{\text{input}} = 3$ and $\text{channel}_{\text{output}} = 64$. $C$ is the number of defined classes in the task. Each conv layer is followed by a BN layer and ReLU function."

The paradigm for our ISS-Net is expressed as follows: pred $= G_\theta(\text{image}, \text{mask}, \text{click})$, where the optional inputs mask $\in \mathbb{R}^{H,W,C}$ and click $\in \mathbb{R}^{H,W,C}$ represent the previous mask and the interactive feature maps obtained through disk encoding [35], respectively. When both mask and click are empty, ISS-Net will perform automatic semantic segmentation. Otherwise, ISS-Net will refine the prediction based on the interactive information.

When constructing ISS-Net, we retain the architecture of mainstream automatic semantic segmentation models and introduce an additional interactive branch to handle the inputs mask and click. This design ensures that our ISS-Net remains compatible with various semantic segmentation models, including transformers [39], [40], [41] and CNNs [42], [43], [44]. To make fair comparisons, we employ ResNet101 [45] as the backbone and Deeplabv2 [46] or Deeplabv3plus [47] as the decoder, which are commonly used in the ADA semantic segmentation. As shown in Fig. 2, we use three convolutional layers to embed the inputs mask and click into an interactive feature map, which is then fused with the image feature map after the first convolutional layer of ResNet101. The rest parts of the ISS-Net are consistent with either Deeplabv3plus-ResNet101 or Deeplabv2-ResNet101. We initialized the empty optional inputs as zero matrixes.

Besides the modifications in the model structure, we introduce interactive semantic segmentation learning to equip ISS-Net with the capability of interactive refinement, as shown in Fig. 1(c). Before entering the DA stage, we conduct this learning process based on labeled source data. We utilize interactive predictions $p^s_{\text{iseg}}$ and ground truth $y^s$ to calculate the cross-entropy loss, denoted as $L^s_{\text{iseg}}$, aiming to improve the prediction accuracy during interactive semantic segmentation learning.

In addition, we design a click loss to prompt the model to make predictions based on the interactive information (mask and click). It can be calculated as follows:

$$L_{\text{click}} = \frac{1}{\sum_{i,j,c}^{H,W,C} \text{click}_{i,j,c}} \sum_{i,j}^{H,W} \sum_{c}^{C} \text{click}_{i,j,c} \log\left(p_{\text{iseg}_{i,j,c}}\right) \tag{10}$$

where the click of the source domain is sampled based on the corrective click sampler [18], [35] and $p_{\text{iseg}}$ is the interactive prediction with mask or click input. The corrective click sampler automatically places clicks with the correct labels near the center of the largest mislabeled region in the prediction.

The click loss generated by the source data is represented as $L^s_{\text{click}}$, and its weight is denoted as $\lambda^s_{\text{click}}$.

### *E. Pseudo-Labels Retraining With Consistency Regularization*

In the DA stage, we generate dense and accurate pseudo-labels through the interactive refinement capability of ISS-Net. Subsequently, we enhance the performance in the target domain through retraining and consistency regularization.

*1) Pseudo-Labels Retraining:* We treat the target label $Y^t_{\text{point}}$ as click inputs to ISS-Net and obtain the refined prediction $p^t_{\text{iseg}}$, as shown in Fig. 1(d). Then, the pseudo-labels, denoted as $y^t_{\text{pseudo}}$, are generated base on the entropy of each pixel in $p^t_{\text{iseg}}$, as shown below

$$y^t_{\text{pseudo}_{i,j}} = \begin{cases} \arg\max_c \left(p^t_{\text{iseg}_{i,j}}\right), & u^t_{\text{iseg}_{i,j}} < \gamma_p \\ 255, & \text{otherwise} \end{cases} \tag{11}$$

where $u^t_{\text{iseg}_{i,j}}$ is the entropy [see (8)] of pixel $(i, j)$ in the target prediction $p^t_{\text{iseg}}$ and $\gamma_p$ is the entropy threshold.

As shown in Fig. 1(d), to achieve DA for ISS-Net, we employ the cross-entropy loss to compute the target pseudo-label loss $L^t_{\text{pseudo}}(p^t, y^t_{\text{pseudo}})$ and the target click loss $L^t_{\text{click}}(p^t_{\text{iseg}}, y^t_{\text{point}})$. The weight of the target pseudo-label loss and the target click loss are denoted as $\lambda^t_{\text{pseudo}}$ and $\lambda^t_{\text{click}}$, respectively.

*2) Consistency Regularization:* The sparsity of target labels and the inaccuracy of pseudo-labels introduce noise to DA. To mitigate this issue, we introduce consistency regularization, as shown in Fig. 3. Overall, four feedforward streams are maintained for each target mini-batch: i) the feature-level perturbation stream: $x^t_w \xrightarrow{\text{FP}} p^t_{\text{fp}}$. ii) and iii) two separate image-level strong perturbation streams: $x^t_{s1} \rightarrow p^t_{s1}$, $x^t_{s2} \rightarrow p^t_{s2}$. iv) the pesudo-labels generation stream: $(x^t_w, p^t_w, y^t_{\text{point}}) \rightarrow p^t_{w_{\text{iseg}}}$. In all streams, we first apply weak augmentation to the target images $x^t$ by randomly resizing them with a scale ranging from 0.5 to 2.0, which helps align the domain gaps in spatial resolution. In the image-level strong perturbation stream, we additionally apply color transformations from ST++ [48] and CutMix [49], while implementing a 50% probability channel Dropout [24] in the feature-level perturbation stream. By aligning the outputs of stream i) with stream iv), as well as the outputs of stream

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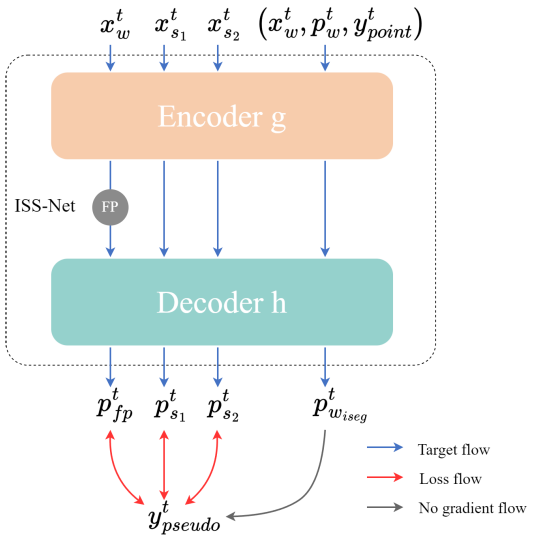



Fig. 3. Consistency regularization with our ISS-Net. "FP" denotes feature perturbations, which is a simple Dropout. The subscripts "$w$" and "$s$" indicate weak and strong data augmentation, respectively. $p_w^t$ represents the target prediction with weak augmentation but without feature perturbations.

ii) and iii) with stream iv), our ISS-Net can adapt to various appearance characteristics arising from spectral channels and data noises in the target labels, thereby learning more robust features from the target domain. Consequently, besides the three losses mentioned in Section III-E1, two cross-entropy losses, $L_{\text{fp}}^t$ for the stream i) and $L_s^t$ for the stream ii) and iii), are introduced to achieve multilevel consistency for our ISS-Net. The weights of loss $L_{\text{fp}}^t$ and $L_s^t$ are denoted as $\lambda_{\text{fp}}^t$ and $\lambda_s^t$, respectively, and are both set to 0.5 as in [24].

### F. Training Process

In the implementation, our detailed training process is shown in Algorithm 1. The four inputs in the algorithm are labeled source data $(X^s, Y^s)$, unlabeled target data $(X^t, Y_{\text{point}}^t)$, total iteration number $\text{Iter}_{\text{total}}$, and labeling rounds $\text{Iters}_{\text{label}}$. The output of the algorithm is the optimized model $G_\theta$. Before labeling the target data, we utilize in Fig. 1(a) and (c) for warm-up training. In Fig. 1(a), we obtain source prediction $p_0^s$ and target prediction $p_0^t$ based on the automatic semantic segmentation ability of ISS-Net $G_\theta$, where the subscript "0" indicates automatic semantic segmentation without interactive inputs. In Fig. 1(b), we first randomly choose the number of clicks, denoted as $N$, between 1 and $N_{\text{click}}$. Then, we refine the source prediction $p_0^s$ step by step, where $N_{\text{click}}$ is the maximum number of clicks and is set to 40 by default. The final interactive prediction of source data is denoted as $\text{pred}_N^s$, where the subscript "N" indicates that the prediction has been refined by $N$ clicks. As described in Sections III-B and III-D, the total loss $L_{\text{total}}$ for the warm-up training is shown in line 18 of Algorithm 1.

In specific rounds, we label the target data via SFAL as shown in lines 39–42 of Algorithm 1. Then, lines 21–36 of Algorithm 1 are added to the training, i.e., the model steps into the DA training phase. The loss $L_{\text{seg}}^t$ in line 21 represents the target semantic segmentation loss in Fig. 1(a). The target label $y_{\text{point}}^t$ is regarded as the input click to ISS-Net and then

**Algorithm 1** Training Process

**Input:** labeled source data $(X^s, Y^s)$, unlabeled target data $(X^t, \varnothing)$, total iteration number $Iter_{total}$, labeling rounds $Iters_{label} = [iter_1, iter_2, \ldots, iter_k]$
**Output:** optimized $G_\theta$

1 $Y_{point}^t = \varnothing$
2 **for** *iter* $= 0$ *to* $Iter_{total}$ **do**
3  # Domain adaptation semantic segmentation learning
4  $x^s, y^s = fetch(X^s, Y^s)$
5  $x^t, y_{point}^t = fetch(X^t, Y_{point}^t)$
6  $p_0^s = G_\theta(x^s, None, None)$
7  $L_{seg}^s = L_{seg}(p_0^s, y^s)$ # Eq. 2
8  $p_0^t = G_\theta(x^t, None, None)$
9  $L_{nl}^t = L_{nl}(p_0^t)$ # Eq. 4
10  # Interactive Semantic Segmentation learning
11  $N = random(1, N_{thres})$ # random click number
12  **for** $j = 1$ *to* $N_{click}$ **do**
13  $click = ClickSampler(pred_{j-1}^s, y^s)$
14  $pred_j^s = G_\theta(x^s, pred_{j-1}^s, click)$
15  **end**
16  $L_{iseg}^s = L_{seg}(pred_N^s, y^s)$ # Eq. 2
17  $L_{click}^s = L_{click}(pred_N^s, click)$ # Eq. 10
18  $L_{total} = L_{seg}^s + L_{iseg}^s + \lambda_{click}^s L_{click}^s + \lambda_{nl}^t L_{nl}^t$
19  # Pseudo-labels retraining with consistency regularization
20  **if** $y_{point}^t$ *is not* $\varnothing$ **then**
21  $L_{seg}^t = L_{seg}(p_0^t, y_{point}^t)$ # Eq. 2
22  # Pseudo-labels retraining
23  # Stream (iv)
24  $p_{w_{iseg}}^t = G_\theta(x^t, p_0^t, y_{point}^t)$
25  $y_{pseudo}^t = PseudoLabelGenerator(p_{w_{iseg}}^t)$ # Eq. 11
26  $L_{pseudo}^t = L_{seg}(p_0^t, y_{pseudo}^t)$ # Eq. 2
27  $L_{click}^t = L_{click}(p_{w_{iseg}}^t, y_{point}^t)$ # Eq. 10
28  # Consistency Regularization
29  # Stream (i)
30  $p_{fp}^t = G_\theta(x^t, None, None, fp = True)$
31  $L_{fp}^t = L_{seg}(p_{fp}^t, y_{pseudo}^t)$
32  # Stream (ii) and (iii)
33  $x_{s_1}^t, x_{s_2}^t = DataAug(x^t)$
34  $p_{s_1}^t, p_{s_2}^t = G_\theta(Cat(x_{s_1}^t, x_{s_2}^t), None, None).chunk(2)$
35  $L_s^t = L_{seg}(p_{s_1}^t, y_{pseudo}^t) + L_{seg}(p_{s_2}^t, y_{pseudo}^t)$ # Eq. 2
36  $L_{total} = L_{total} + L_{seg}^t + \lambda_{pseudo}^t L_{pseudo}^t + \lambda_{click}^t L_{click}^t + \lambda_{fp}^t L_{fp}^t + \lambda_s^t L_s^t$
37  **end**
38  Update $G_\theta$ by $L_{total}$
39  # Image Labeling via SFAL
40  **if** *iter in* $Iters_{label}$ **then**
41  Update $Y_{point}^t$ by SFAL
42  **end**
43 **end**

we obtain the refined target prediction $p_{w_{\text{iseg}}}^t$. Subsequently, we produce the pseudo-labels based on (11), and calculate the additional loss in Fig. 1(d) as mentioned in Section III-E. The total loss of the DA training phase is shown in line 36 in Algorithm 1.

## IV. Experiment

### A. Data Description

To validate our method, we conduct experiments using three datasets: Potsdam [50], Vaihingen [51], and Semantic Drone [52]. Based on the dataset sizes, we consider the Potsdam dataset as the source data, while treating the

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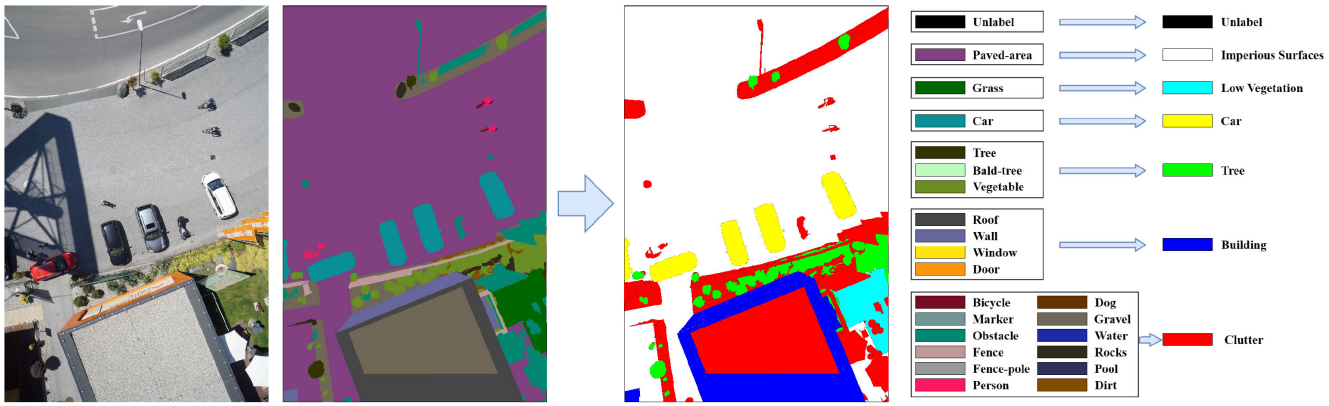



Fig. 4. Label adjustment process for the Semantic Drone dataset.

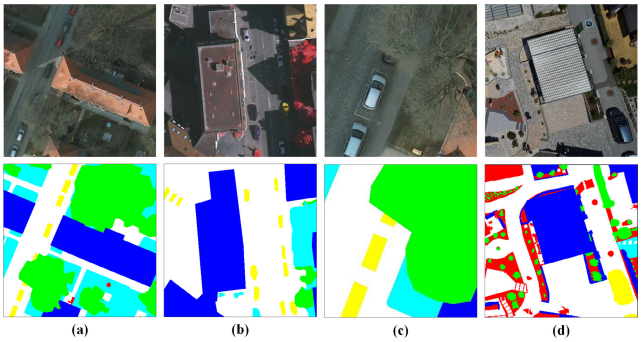



Fig. 5. Potsdam, Vaihingen, and Semantic Drone image patches with different resolutions. (a) Potsdam RGB of size 960 × 960 pixels, (b) Vaihingen IRRG of size 512 × 512 pixels, (c) Potsdam RGB of size 320 × 320 pixels, and (d) Semantic Drone RGB of size 320 × 320 pixels.

Vaihingen and Semantic Drone datasets as the target data. Potsdam and Vaihingen, provided by the International Society for Photogrammetry and Remote Sensing (ISPRS), are both aerial remote sensing image datasets. The Semantic Drone dataset, provided by the Institute of Computer Graphics and Vision (ICG), is an unmanned aerial vehicle (UAV) remote sensing dataset. The Potsdam dataset contains 38 tiles with a spatial resolution of 5 cm and a size of 6000 × 6000 pixels, each consisting of a true orthophoto (TOP). Except for one erroneous mask, the remaining 37 patches have been manually pixel-wise labeled into six classes: building, impervious surfaces, tree, low vegetable, car, and clutter. We choose the red, green, and blue (RGB) channel composition of the Potsdam dataset for experiments. The Vaihingen dataset is composed of 33 tiles with a spatial resolution of 9 cm and size near 2000 × 2000 pixels. All these tiles have the same six classes as the Potsdam dataset and the channel composition of InfraRed, red, and green (IRRG). The Semantic Drone dataset contains 400 RGB images with a size of 6000 × 4000 pixels captured by a high-resolution camera at altitudes ranging from 5 to 30 m above the ground. For the semantic segmentation task, it provides pixel-wise annotations with 23 classes, such as grass, car, tree, bald-tree, vegetable, roof, wall, and so on, as shown in Fig. 4.

*1) Potsdam → Vaihingen:* The domain gaps for the Potsdam → Vaihingen task lie in spatial resolution, channel composition, and appearance characteristics. To align the spatial resolution, we follow [5] to crop the Potsdam and Vaihingen tiles into patches sized 960 × 960 and 512 × 512, respectively, as shown in Fig. 5. A total of 4093 patches are cropped from 37 Potsdam tiles, serving as the training data for the source domain. Similarly, 1784 patches are cropped from 26 Vaihingen tiles (tile IDs: {1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22, 23, 24, 26, 27, 28, 29, 30}) and utilized as the training data for the target domain. The remaining seven Vaihingen tiles (tile IDs: {31, 32, 33, 34, 35, 37, 38}) are cropped into 1784 patches to serve as the test set for the target domain.

*2) Potsdam → Semantic Drone:* The domain gaps for the Potsdam → Semantic Drone task include differences in spatial resolution, appearance characteristics, and object diversity. The varying capture altitudes in the Semantic Drone dataset make it unfeasible to align the spatial resolution through data preprocessing. However, UAVs fly at lower altitudes, experience less atmospheric disturbance, exhibit fewer image blurriness and distortions, and capture richer details, contributing to the high quality of the Semantic Drone dataset. We first resize the Semantic Drone data to 600 × 400 and utilize 320 × 320 windows to sample patches from both the Potsdam and Semantic Drone datasets, resulting in 11 988 training samples for the source domain and 1600 samples for the target domain. Within the target domain samples, we allocate 1200 for training and 400 for testing. To align the categories in the Semantic Drone dataset with those in the Potsdam dataset, we make adjustments as illustrated in Fig. 4. As shown in Fig. 4, the Semantic Drone dataset contains various objects that are not present in the Potsdam dataset, leading to a significant challenge for this DA task.

### B. Evaluation Metrics

To make a comprehensive and unbiased evaluation, we adapt the mean intersection of union (mIoU), $F1$ score, mean accuracy (MA), and overall accuracy (OA) metrics, which are commonly used in semantic segmentation.

mIoU assesses the degree of overlap between the prediction and ground truth mask, providing insights into the segmentation quality overall. It can be calculated as follows:

$$\text{mIoU} = \frac{1}{C}\sum_{c}^{C}\frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} \tag{12}$$

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where $TP_c$, $FP_c$, and $FN_c$ represent the true positive, false positive, and false negative of class $c$, respectively.

The $F1$ score is a measure that balances precision and recall. In semantic segmentation, it quantifies how well the model correctly identifies positive pixels (true positives) while minimizing false positives (false positives) and false negatives (false negatives). It can be calculated as follows:

$$F1 = \frac{1}{C}\sum_{c}^{C}\frac{2\cdot \mathrm{TP}_c}{2\cdot \mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}. \tag{13}$$

MA calculates the average accuracy for each class and provides insights into how well the model performs for each class. It can be calculated as follows:

$$\mathrm{MA} = \frac{1}{C}\sum_{c}^{C}\frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}. \tag{14}$$

OA measures the proportion of correctly classified pixels over the entire dataset. It can be computed as follows:

$$\mathrm{OA} = \frac{\sum_c^C \mathrm{TP}_c}{\sum_c^C \mathrm{TP}_c + \sum_c^C \mathrm{FN}_c}. \tag{15}$$

Higher values of all these metrics indicate better segmentation performance.

### C. Experimental Settings

All subsequent experiments are conducted using PyTorch on a single NVIDIA GeForce RTX 3090 GPU with 24 GB RAM. We employ the stochastic gradient descent (SGD) optimizer with a momentum of 0.9 to train our model and set the batch size to 2. The initial learning rate of the pretrained ResNet-101 in the encoder is set to $2.5 \times 10^{-4}$, while the initial learning rate of the interaction branch and decoder is set to $2.5 \times 10^{-3}$. The learning rate is decayed using a polynomial decay strategy with a coefficient of 0.9. Our model is trained for a total of 25 000 iterations, and the labeling rounds are set at [3000, 6000, 9000, 12 000, 15 000], indicating 3000 iterations of warm-up training. By default, the weights $\lambda^s_{\text{click}}$ and $\lambda^t_{\text{click}}$ are both set to 2.0 and the entropy threshold $\gamma_p$ is set to 0.05. In the comparative experiments, the weight $\lambda^t_{\text{pseudo}}$ and error threshold $\epsilon_{\text{error}}$ are set to 0.8 and 0.3, respectively.

### D. Benchmarking

To fully validate the effectiveness and advantages of our method, we select several state-of-the-art methods to conduct experiments. We compare the proposed method with five existing (UDA) methods, namely, ADVENT [53], AdaptSegNet [29], CLAN [28], UDAS [5], and MBATA-GAN [12], as well as four advanced ADA methods, namely, MADAv2 [54], ILM-ASSL [9], D2ADA [23], and RIPU [18].

1) ADVENT [53] introduces an entropy loss to penalize uncertain predictions in the target domain and proposes an entropy-based adversarial training approach to align entropy maps between the source and target domains.
2) AdaptSegNet [29] employs adversarial learning to achieve multilevel alignment across domains.
3) CLAN [28] proposes category-level adversarial alignment with dynamically weighted adversarial loss to enhance feature alignment in DA.
4) UDAS [5] leverages category-certainty attention and category-level adversarial loss to enhance cross-domain semantic segmentation for high-resolution satellite imagery.
5) MBATA-GAN [12] leverages MBAtrans for semantic alignment via cross-attention and self-attention and introduces a multidiscriminator module to learn both domain-invariant and domain-specific representations simultaneously.
6) MADAv2 [54], an improved version of MADA [55], favors selecting target domain outliers that complement the source domain and employs a robust semi-supervised strategy to learn from samples near the target anchors.
7) ILM-ASSL [9] computes the average entropy of each prediction to obtain an image uncertainty score for selecting target samples. It enhances segmentation performance by implementing a series of semi-supervised learning strategies, namely, U2PL [56] and AEL [34].
8) D2ADA [23] partitions target domain images into regions using superpixels, selects image regions for annotation based on domain density, and introduces a dynamic scheduling policy to adjust resource allocation between model uncertainty and domain exploration.
9) RIPU [18] leverages region impurity and prediction uncertainty to identify image regions that exhibit both spatial diversity and prediction uncertainty.

It's worth noting that MADAv2 and ILM-ASSL focus on image-level active learning by annotating a subset of target domain images, while D2ADA adopts a region-level approach, annotating informative regions across the target domain. Meanwhile, both RIPU and our method employ point-level active learning, acquiring information-rich pixels across the target domain. The labeling cost for all the above methods is calculated as the proportion of annotated pixels in the target domain relative to the total number of pixels. For a fair comparison, we control the labeling cost of image-level and region-level methods to 1%, i.e., 1% of the pixels get the ground truths from humans. To highlight the advantages of our method, both point-level methods label only 40 pixels per image. In other words, our labeling cost is $40/(512 \times 512) = 0.015\%$ for the Potsdam $\rightarrow$ Vaihingen task, and $40/(320 \times 320) = 0.039\%$ for the Potsdam $\rightarrow$ Semantic Drone task. In addition, we include two control groups: one trained on source data without DA (no adaptation) and another trained on target data with full supervision (fully supervised) for comparison.

### E. Experimental Result Analysis

*1) Comparative Studies on Potsdam $\rightarrow$ Vaihingen:* From Table I, the model without adaptation shows the poorest performance due to domain shifts, achieving 48.88% OA, 48.18% MA, and 25.93% mIoU. UDA methods can improve segmentation performance; for example, CLAN shows the best performance of 65.69%, 58.39%, and 39.99% for OA, MA, and mIoU, respectively. However, the performance of

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TABLE I
COMPARISON RESULTS ON THE ADAPTATION TASK POTSDAM → VAIHINGEN (BOLD: BEST)

| Method | Labeling cost | Building | Imp. surf. | Tree | Low veg. | Car | Clutter | OA | MA | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|
| No Adaptation | 0% | 59.38 | 52.12 | 60.69 | 44.36 | 2.74 | 8.92 | 48.88 | 48.18 | 25.93 |
| ADVENT | 0% | 72.31 | 70.00 | 65.95 | 33.52 | 39.17 | 15.54 | 60.49 | 50.69 | 34.99 |
| AdaptSegNet | 0% | 73.75 | 72.98 | 66.00 | 27.25 | 50.77 | 11.16 | 62.18 | 54.14 | 36.86 |
| CLAN | 0% | 76.38 | 77.58 | 70.99 | 37.20 | 44.99 | 14.83 | 65.69 | 58.39 | 39.99 |
| UDAS | 0% | 80.11 | 72.05 | 54.13 | 57.30 | 33.08 | 16.34 | 65.48 | 53.27 | 38.18 |
| MBATA-GAN‡ | 0% | 80.97 | 73.03 | 49.92 | 53.49 | 4.75 | 30.50 | 65.86 | 48.50 | 35.96 |
| D2ADA | 1% | 93.00 | 88.57 | 77.67 | 78.27 | **72.16** | 52.78 | 84.91 | 75.19 | 64.41 |
| RIPU | 0.015% | 92.15 | 88.10 | 76.57 | 78.84 | 67.23 | 30.44 | 84.36 | 70.22 | 59.98 |
| Ours | 0.015% | **93.20** | **89.39** | **80.97** | **80.74** | 70.83 | **60.78** | **86.35** | **78.98** | **67.05** |
| Full Supervised | 100% | 91.99 | 89.54 | 79.41 | 81.71 | 68.37 | 54.59 | 86.08 | 75.01 | 65.11 |
| MADAv2† | 1% | 74.95 | 79.32 | 70.35 | 68.06 | 35.48 | 24.76 | 73.31 | 57.00 | 44.53 |
| ILM-ASSL† | 1% | 93.21 | 89.52 | 78.32 | 76.79 | 72.72 | **66.67** | 85.08 | 78.68 | 67.02 |
| D2ADA† | 1% | 93.40 | 89.33 | 80.02 | 79.64 | 73.08 | 55.58 | 86.02 | 77.12 | 66.21 |
| RIPU† | 0.015% | 92.57 | 88.94 | 78.80 | 78.98 | 69.30 | 47.54 | 85.13 | 74.35 | 63.46 |
| Ours† | 0.015% | **93.74** | **89.88** | **81.77** | **80.42** | **73.34** | 63.38 | **86.82** | **80.14** | **68.42** |
| Full Supervised† | 100% | 92.17 | 89.17 | 81.23 | 81.50 | 70.96 | 62.25 | 86.41 | 77.83 | 67.21 |

Methods with † and ‡ are based on Deeplabv3plus-ResNet101 and Transformer, respectively, while others are based on Deeplabv2-ResNet101.

UDA methods still lags far behind the fully supervised model, with OA, MA, and mIoU of 86.08%, 75.01%, and 65.11%, respectively. By labeling a small amount of target data, ADA methods can effectively bridge domain gaps and significantly improve the model's performance. MADAv2, ILM-ASSL, and D2ADA all employ image-level or region-level labeling strategies, densely labeling either the entire image or specific regions. Recent works, D2ADA and ILM-ASSL, show strong performance, approaching the fully supervised model's performance at a 1% labeling cost. The Deeplabv3plus-ResNet101-based (V3-based) ILM-ASSL achieves only a 0.19% lower mIoU and 1.33% lower OA compared to the fully supervised model. Moreover, it even surpasses the fully supervised model by 0.85% in terms of MA. Since dense labels can directly provide contextual information about object boundaries at a higher labeling cost, ILM-ASSL and D2ADA outperform point-level RIPU significantly. In contrast, our point-level method achieves the best performance with only a 0.015% labeling cost, surpassing the fully supervised model in terms of OA, MA, and mIoU. Based on Deeplabv2-ResNet101 (V2-based), our approach outperforms D2ADA by 0.97%, 1.95%, and 1.32% in terms of OA, MA, and mIoU, respectively. Similarly, using Deeplabv3plus-ResNet101, our method surpasses ILM-ASSL by 1.74%, 1.46%, and 1.40% in OA, MA, and mIoU, respectively. Furthermore, our V2-based model also achieves the best results in OA, MA, and mIoU metrics, compared to other V3-based methods. These substantial performance improvements demonstrate the significant advantages of our method in the Potsdam → Vaihingen task.

In Table I, our method achieves competitive $F1$ scores across all categories. Specifically, our V2-based method exhibits improvements of 0.20%, 0.82%, 3.30%, 1.90%, and 8.00% compared to the runner-up in building, impervious surfaces, tree, low vegetation, and clutter categories, respectively. In addition, our V3-based method improves by 0.34%, 0.36%, 1.75%, 0.78%, and 0.26% over the second-place method in the building, impervious surfaces, tree, low vegetation, and car categories, respectively. Furthermore, our method even outperforms the fully supervised model in most categories, with noteworthy improvements, such as 1.57%, 0.71%, 0.54%, 2.38%, and 1.13% in the building, impervious surfaces, tree, car, and clutter categories for V3-based model, respectively. We attribute the superior performance in specific classes of image-level and region-level methods to their ability to provide rich contextual information. By addressing the class imbalance problem through AEL [34], ILM-ASSL performs well on the rare class Clutter. However, the extensive labeling budget wasted on redundant areas results in poor performance for ILM-ASSL in the tree and low vegetation categories.

In Fig. 7, ILM-ASSL exhibits substantial confusion in the tree and low vegetation classes, while D2ADA performs poorly in the clutter category. In contrast, our method consistently outperforms other approaches, including the fully supervised model. These competitive results demonstrate that our method has the potential to significantly enhance the performance of point-level methods, highlighting the robust potential of point-level approaches.

*2) Comparative Studies on Potsdam → Semantic Drone:* To further demonstrate the effectiveness of our method, we conduct comparative experiments on a more challenging task: Potsdam → Semantic Drone. As shown in Table II, due to the high quality of the Semantic Drone dataset, most methods outperform their performance on the Vaihingen target domain. Similarly, UAD methods can improve performance in the target domain, compared to the model without adaptation. However, due to the substantial domain gap, they achieve only about 50% of the performance of the fully supervised model, which is lower than the 60% achieved on the Potsdam → Vaihingen task. Due to the object diversity in the Semantic Drone dataset, the images selected by MADAv2 and ILM-ASSL do not provide sufficient supervised information, resulting in their poor performance. For instance, the V3-based

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TABLE II
COMPARISON RESULTS ON THE ADAPTATION TASK POTSDAM → SEMANTIC DRONE (BOLD: BEST)

| Method | Labeling cost | Building | Imp. surf. | Tree | Low veg. | Car | Clutter | OA | MA | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|
| No Adaptation | 0% | 49.69 | 63.82 | 4.68 | 79.11 | 0.00 | 0.00 | 56.60 | 34.61 | 24.68 |
| ADVENT | 0% | 31.43 | 59.49 | 39.98 | 53.22 | 42.67 | 18.75 | 62.02 | 57.40 | 40.92 |
| AdaptSegNet | 0% | 56.66 | 76.20 | 56.90 | 56.83 | 61.65 | 19.62 | 59.71 | 55.86 | 39.41 |
| CLAN | 0% | 49.68 | 70.91 | 42.70 | 40.20 | 69.35 | 35.79 | 51.78 | 50.54 | 35.93 |
| UDAS | 0% | 57.48 | 76.50 | 59.22 | 79.42 | 47.88 | 22.42 | 66.44 | 57.98 | 42.39 |
| MBATA-GAN‡ | 0% | 62.86 | 77.20 | 62.98 | 68.62 | 47.78 | 14.75 | 64.12 | 62.98 | 41.04 |
| D2ADA | 1% | 77.01 | 89.41 | 78.86 | 92.74 | 91.58 | 70.66 | 84.32 | 83.93 | 72.36 |
| RIPU | 0.039% | 84.13 | 92.22 | 81.35 | 95.71 | 90.62 | 76.74 | 88.27 | 86.62 | 77.45 |
| Ours | 0.039% | **88.19** | **94.45** | **82.96** | **96.06** | **91.61** | **81.63** | **90.70** | **89.31** | **81.06** |
| Full Supervised | 100% | 88.18 | 94.42 | 85.23 | 96.44 | 92.44 | 82.60 | 91.18 | 90.15 | 82.20 |
| MADAv2† | 1% | 64.65 | 76.17 | 59.72 | 80.19 | 36.96 | 45.02 | 68.40 | 66.39 | 45.08 |
| ILM-ASSL† | 1% | 51.20 | 82.19 | 60.52 | 80.62 | 82.90 | 48.93 | 69.58 | 70.59 | 53.15 |
| D2ADA† | 1% | 82.75 | 91.14 | 79.37 | 93.23 | 89.98 | 73.87 | 86.25 | 85.22 | 74.63 |
| RIPU† | 0.039% | 84.63 | 92.63 | 82.67 | 93.95 | 90.80 | 79.76 | 88.84 | 87.39 | 78.21 |
| Ours† | 0.039% | **89.08** | **94.75** | **83.97** | **96.07** | **92.78** | **82.23** | **91.11** | **90.05** | **82.12** |
| Full Supervised† | 100% | 89.55 | 94.68 | 85.55 | 96.43 | 93.64 | 83.30 | 91.56 | 90.72 | 83.24 |

Methods with † and ‡ are based on Deeplabv3plus-ResNet101 and Transformer, respectively, while others are based on Deeplabv2-ResNet101.

ILM-ASSL achieves only 69.58% OA, 70.59% MA, and 53.15% mIoU. In constant, the sparser labeling methods like D2ADA and RIPU yield better results. Notably, the point-level RIPU achieves the second-best performance, with scores of 88.84% OA, 87.39% MA, and 78.21% mIoU. This indicates that sparse annotations can offer richer information to bridge the significant domain gap arising from the dataset's object diversity. Furthermore, our EasySeg outperforms all other methods, achieving OA, MA, and mIoU scores of 91.11%, 90.05%, and 82.12%, respectively, based on the Deeplabv3plus-ResNet101. Due to the significant domain gap, our method does not surpass the fully supervised models in this task. However, when compared to the fully supervised model, our V2-based model lags behind by only 0.48% OA, 0.84% MA, and 1.14% mIoU, while our V3-based model lags behind by only 0.45% OA, 0.67% MA, and 1.12% mIoU.

In Table II, our method outperforms all others in $F1$ scores across all categories, particularly in the Building and Clutter categories. The reason is that in the Semantic Drone dataset, class Wall, including perimeter walls, is transferred into the Building class, which cannot be learned from the Potsdam dataset. In addition, the clutter class in the Semantic Drone dataset encompasses 12 different objects, increasing the difficulty of DA. While the sparser labeling of RIPU and D2ADA partially addresses the challenge of object diversity, their misconceptions in these two classes remain severe. In contrast, our error-aware method further mitigates this noise, enabling the model to achieve results comparable to or even better than the fully supervised model in all categories.

As shown in Fig. 8, other methods exhibit poor prediction quality for building and clutter classes, while our predictions closely align with those of the fully supervised model. These outstanding results strongly demonstrate the superiority of our method.

*3) Ablation Study:* We conduct ablation experiments on the key components of the proposed EasySeg. As shown in Tables III and IV, we set RIPU [18] as the baseline and compare it with our models under different configurations in two adaptation tasks.

TABLE III
ABLATION RESULTS FOR POTSDAM → VAIHINGEN

| Method | OA | MA | mIoU |
|---|---|---|---|
| RIPU | 84.36 | 70.22 | 59.98 |
| RIPU+ISS-Net | 84.37 | 72.31 | 61.85 |
| RIPU+SFAL | 84.60 | 74.89 | 63.18 |
| RIPU+ISS-Net+SFAL | 84.88 | 75.09 | 64.36 |
| RIPU+SFAL+CoRe | 86.23 | 75.88 | 65.43 |
| RIPU+ISS-Net+SFAL+CoRe | **86.35** | **78.98** | **67.05** |
| RIPU† | 85.13 | 74.35 | 63.46 |
| RIPU†+ISS-Net | 85.41 | 75.64 | 64.61 |
| RIPU†+SFAL | 85.17 | 77.13 | 64.48 |
| RIPU†+ISS-Net+SFAL | 85.53 | 76.73 | 65.05 |
| RIPU†+SFAL+CoRe | 86.82 | 76.78 | 67.31 |
| RIPU†+ISS-Net+SFAL+CoRe | **86.82** | **80.14** | **68.42** |

Methods with † are based on Deeplabv3plus-ResNet101, while others are based on Deeplabv2-ResNet101. "CoRe" means consistency regularization.

In the configuration "RIPU + ISS-Net," we replace the segmenter in the baseline with our ISS-Net to generate dense pseudo-labels. The results in Tables III and IV indicate that our V2-based and V3-based models achieve performance improvements in both adaptation tasks. For example, compared to the baseline, our V2-based model improves by 0.01% OA, 2.09% MA, and 1.87% mIoU on Potsdam → Vaihingen task, while our V2-based model achieves a 1.06% improvement in OA, 1.41% in MA, and 1.12% in mIoU on Potsdam → Semantic Drone task. In Tables V and VI, we analyze the impact of the target pseudo-label loss weight, $\lambda^t_{\text{pseudo}}$. We observe performance improvements with the inclusion of pseudo-labels, with the best results obtained at $\lambda^t_p = 0.8$.

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TABLE IV
ABLATION RESULTS FOR POTSDAM → SEMANTIC DRONE

| Method | OA | MA | mIoU |
|---|---|---|---|
| RIPU | 88.27 | 86.62 | 77.45 |
| RIPU+ISS-Net | 89.33 | 88.03 | 78.57 |
| RIPU+SFAL | 89.21 | 88.33 | 78.91 |
| RIPU+ISS-Net+SFAL | 89.63 | 88.55 | 79.22 |
| RIPU+SFAL+CoRe | 90.17 | 88.84 | 80.35 |
| RIPU+ISS-Net+SFAL+CoRe | **90.70** | **89.31** | **81.06** |
| RIPU† | 88.84 | 87.39 | 78.21 |
| RIPU†+ISS-Net | 89.29 | 87.93 | 78.74 |
| RIPU†+SFAL | 89.48 | 87.76 | 79.39 |
| RIPU†+ISS-Net+SFAL | 90.22 | 88.01 | 80.27 |
| RIPU†+SFAL+CoRe | 90.14 | 88.83 | 80.23 |
| RIPU†+ISS-Net+SFAL+CoRe | **91.11** | **90.05** | **82.12** |

Methods with † are based on Deeplabv3plus-ResNet101, while others are based on Deeplabv2-ResNet101. "CoRe" means consistency regularization.

TABLE V
RESULTS OF DIFFERENT $\lambda^t_{\text{pseudo}}$ FOR POTSDAM → VAIHINGEN

| Method | $\lambda^t_{pseudo}$ | OA | MA | mIoU |
|---|---|---|---|---|
| RIPU | 0.0 | 84.36 | 70.22 | 59.98 |
| RIPU+ISS-Net | 0.5 | 84.24 | 70.99 | 61.21 |
| RIPU+ISS-Net | 0.8 | 84.37 | **72.31** | **61.85** |
| RIPU+ISS-Net | 1.0 | **84.78** | 70.67 | 61.65 |
| RIPU† | 0.0 | 85.13 | 74.35 | 63.46 |
| RIPU†+ISS-Net | 0.5 | 84.54 | 76.37 | 64.15 |
| RIPU†+ISS-Net | 0.8 | **85.17** | **77.13** | 64.48 |
| RIPU†+ISS-Net | 1.0 | 84.52 | 74.99 | **64.72** |

Methods with † are based on Deeplabv3plus-ResNet101, while others are based on Deeplabv2-ResNet101.

TABLE VI
RESULTS OF DIFFERENT $\lambda^t_{\text{pseudo}}$ FOR POTSDAM → SEMANTIC DRONE

| Method | $\lambda^t_{pseudo}$ | OA | MA | mIoU |
|---|---|---|---|---|
| RIPU | 0.0 | 88.27 | 86.62 | 77.45 |
| RIPU+ISS-Net | 0.5 | 89.24 | 87.67 | 78.55 |
| RIPU+ISS-Net | 0.8 | **89.33** | **88.03** | **78.57** |
| RIPU+ISS-Net | 1.0 | 88.83 | 87.16 | 77.83 |
| RIPU† | 0.0 | 88.84 | 87.39 | 78.21 |
| RIPU†+ISS-Net | 0.5 | 89.38 | 87.73 | 78.83 |
| RIPU†+ISS-Net | 0.8 | 89.29 | **87.93** | 78.74 |
| RIPU†+ISS-Net | 1.0 | **89.58** | 87.44 | **79.05** |

Methods with † are based on Deeplabv3plus-ResNet101, while others are based on Deeplabv2-ResNet101.

We believe that adding pseudo-labels reduces redundancy in image labeling and improves labeling efficiency.

In the configuration "RIPU + SFAL," we incorporate our SFAL labeling strategy into the baseline. As shown in Tables III and IV, our V2-based model achieves an increase of 0.24% in OA, 4.67% in MA, and 3.50% in mIoU on the Potsdam → Vaihingen task, along with improvements of 0.94% in OA, 1.71% in MA, and 1.46% in mIoU on the Potsdam → Semantic Drone task. This highlights the significance of rectifying substantial yet confident errors for point-level methods. Furthermore, in Tables VII and VIII, we analyze the impact of the error threshold $\epsilon_{\text{error}}$ in the SFAL strategy. It is evident that SFAL is not sensitive to the error threshold, as values ranging from 0.1 to 0.5 all result in performance improvements. This further validates

TABLE VII
RESULTS OF DIFFERENT $\epsilon_{\text{error}}$ FOR POTSDAM → VAIHINGEN

| Method | $\epsilon_{error}$ | OA | MA | mIoU |
|---|---|---|---|---|
| RIPU | 1.0 | 84.36 | 70.22 | 59.98 |
| RIPU+SFAL | 0.5 | **84.77** | 73.86 | 63.37 |
| RIPU+SFAL | 0.4 | 84.72 | 74.36 | **63.72** |
| RIPU+SFAL | 0.3 | 84.60 | 74.89 | 63.18 |
| RIPU+SFAL | 0.2 | 84.72 | **77.45** | 63.59 |
| RIPU+SFAL | 0.1 | 84.31 | 75.83 | 62.97 |
| RIPU | 1.0 | 85.13 | 74.35 | 63.46 |
| RIPU†+SFAL | 0.5 | **85.56** | 74.68 | 64.37 |
| RIPU†+SFAL | 0.4 | 85.27 | 75.45 | 64.36 |
| RIPU†+SFAL | 0.3 | 85.17 | 77.13 | 64.48 |
| RIPU†+SFAL | 0.2 | 85.27 | 76.30 | **64.72** |
| RIPU†+SFAL | 0.1 | 84.25 | **78.27** | 63.94 |

Methods with † are based on Deeplabv3plus-ResNet101, while others are based on Deeplabv2-ResNet101.

TABLE VIII
RESULTS OF DIFFERENT $\epsilon_{\text{error}}$ FOR POTSDAM → SEMANTIC DRONE

| Method | $\epsilon_{error}$ | OA | MA | mIoU |
|---|---|---|---|---|
| RIPU | 1.0 | 88.27 | 86.62 | 77.45 |
| RIPU+SFAL | 0.5 | 88.65 | 87.19 | 78.23 |
| RIPU+SFAL | 0.4 | 88.63 | 87.22 | 78.25 |
| RIPU+SFAL | 0.3 | **89.21** | **88.33** | **78.91** |
| RIPU+SFAL | 0.2 | 88.90 | 87.67 | 78.38 |
| RIPU+SFAL | 0.1 | 89.01 | 88.07 | 78.85 |
| RIPU† | 1.0 | 88.84 | 87.39 | 78.21 |
| RIPU†+SFAL | 0.5 | 89.02 | 87.82 | 78.92 |
| RIPU†+SFAL | 0.4 | **89.67** | **88.45** | **80.04** |
| RIPU†+SFAL | 0.3 | 89.48 | 87.76 | 79.39 |
| RIPU†+SFAL | 0.2 | 89.55 | 88.35 | 79.68 |
| RIPU†+SFAL | 0.1 | 89.61 | 88.06 | 79.42 |

Methods with † are based on Deeplabv3plus-ResNet101, while others are based on Deeplabv2-ResNet101.

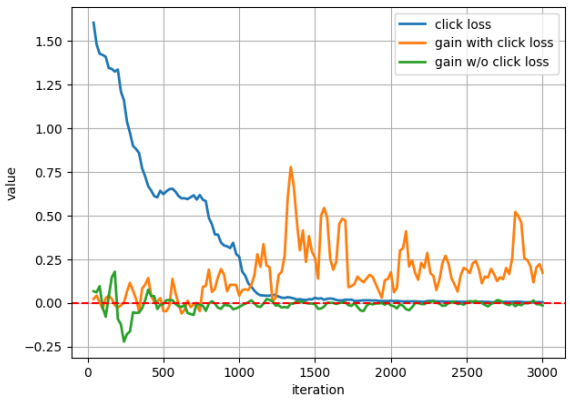



Fig. 6. Gain in mIoU of interactive outputs of the ISS-Net compared to its automatic predictions during warm-up training stage with or without (w/o) click loss.

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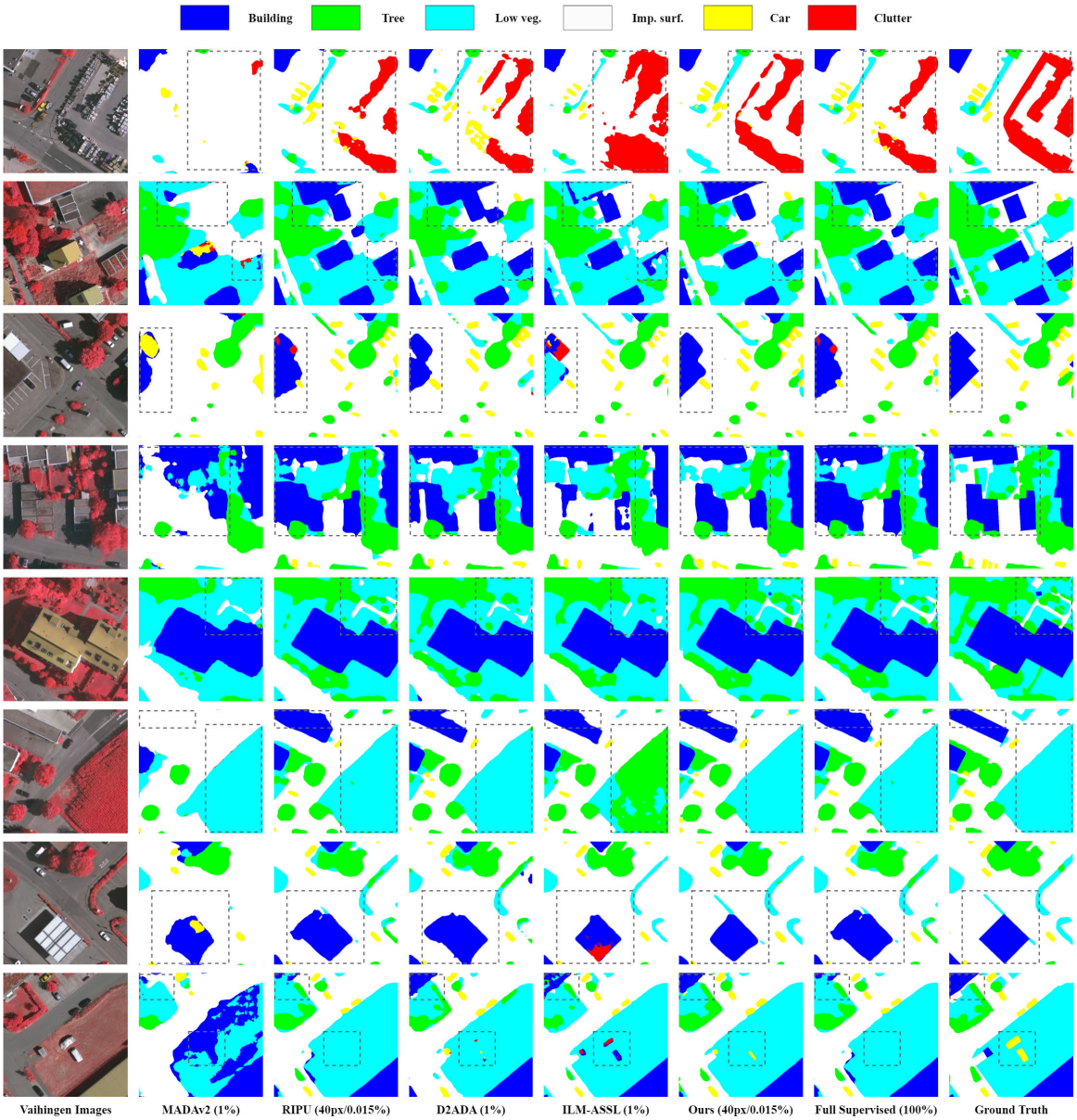



Fig. 7. Qualitative results of Potsdam → Vaihingen task for Deeplabv3plus-ResNet101 based methods.

the necessity of rectifying substantial yet confident errors. In addition, models under configuration "RIPU + SFAL" performed better when the error threshold was set between 0.2 and 0.4, indicating that "See" and "Ask" in our SFAL are two complementary strategies.

In the configuration "RIPU + ISS-Net + SFAL," we incorporated ISS-Net and SFAL into the baseline. The results in Tables III and IV demonstrate that the configuration "RIPU + ISS-Net + SFAL" outperforms both "RIPU + ISS-Net" and "RIPU + SFAL" configurations, highlighting the advantages of combining ISS-Net and SFAL for enhancing DA performance.

In the configuration "RIPU + SFAL + Consistency regularization," we introduce SFAL and consistency regularization into the baseline. Due to the absence of ISS-Net in this configuration, we follow the work [24] to generate pseudo-labels from the prediction $p_w^t$ of the simplest stream. In Tables III and IV, the configuration "RIPU + SFAL + Consistency regularization" demonstrates significant improvements compared to the configuration "RIPU + SFAL." For example, compared to the configuration "RIPU + SFAL," our V2-based model under the configuration "RIPU + SFAL + Consistency regularization" achieves a 1.63% increase in OA, 0.99% in MA, and 2.25% in mIoU on the Potsdam → Vaihingen task, as well as a 0.96% increase in OA, 0.51% in MA, and 1.44% in mIoU on the Potsdam → Semantic Drone task. These results highlight the effectiveness of using consistency regularization to make better use of unlabeled target data, enabling the model to learn more robust features from the target domain.

For the configuration "RIPU + ISS-Net + SFAL + Consistency regularization," we add all our components to the baseline to form our complete EasySeg. The results show that the configuration "RIPU + ISS-Net + SFAL + Consistency regularization" outperforms the configuration "RIPU + SFAL + Consistency regularization," indicating that the

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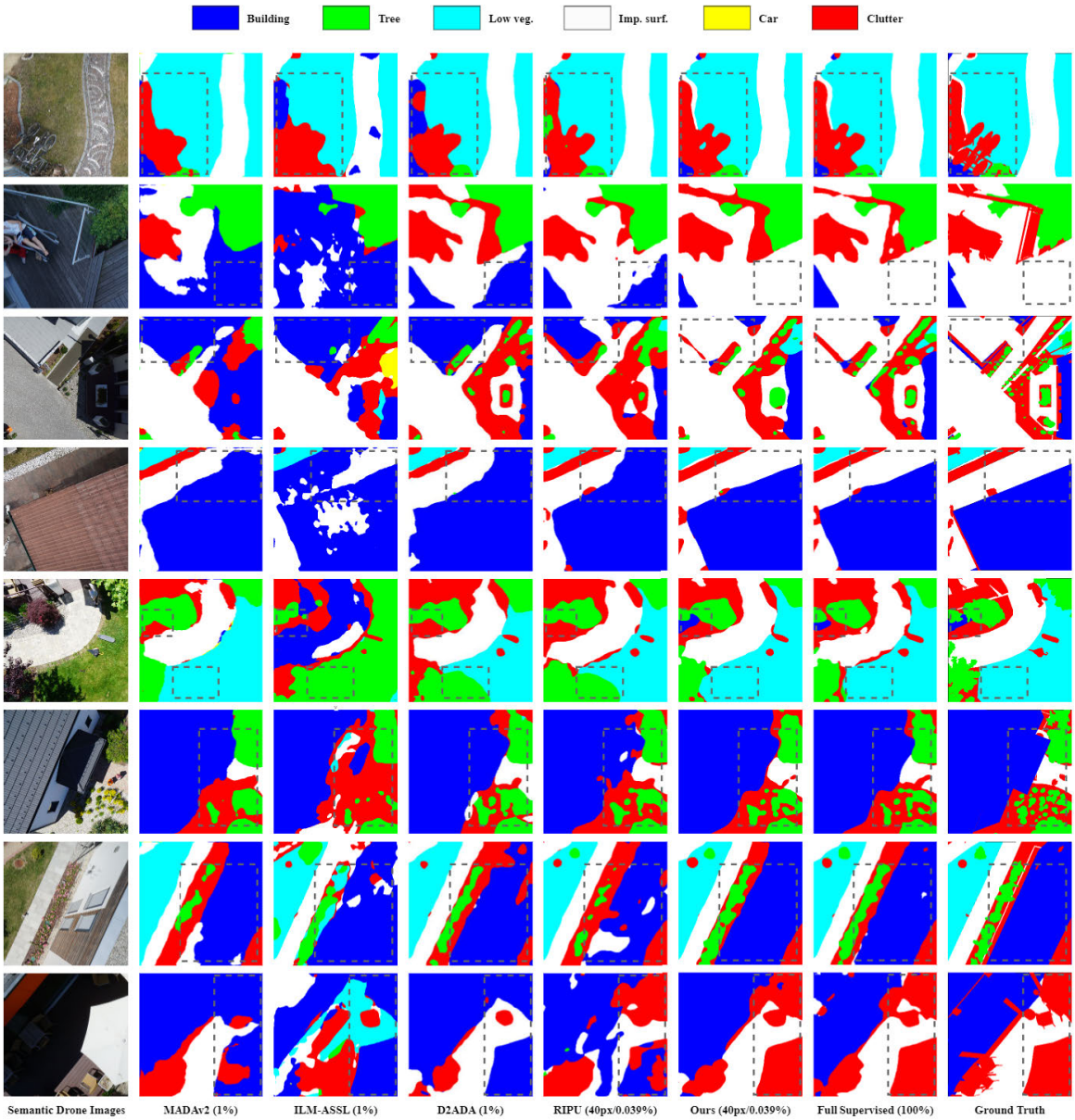



Fig. 8. Qualitative results of Potsdam → Semantic Drone task Deeplabv3plus-ResNet101 based methods.

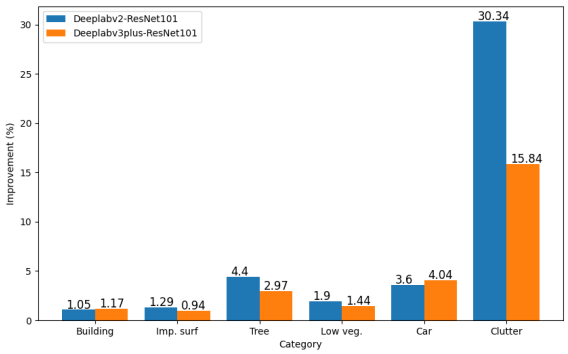



Fig. 9. $F1$ score improvement in each class for our method compared to the baseline RIPU on the Potsdam → Vaihingen task.

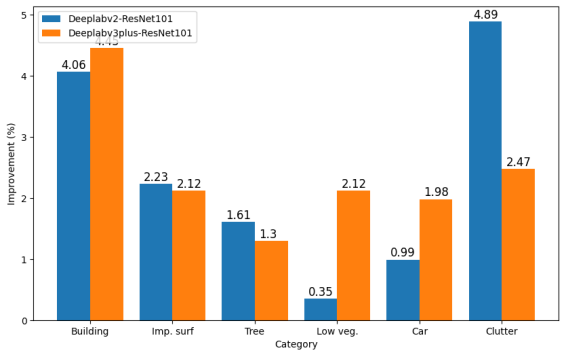



Fig. 10. $F1$ score improvement in each class for our method compared to the baseline RIPU on the Potsdam → Semantic Drone task.

pseudo-labels generated by ISS-Net are more accurate and beneficial for mitigating noise in ADA.

*4) Indispensability of Click Loss:* In this section, we illustrate the essential role of click loss in successfully constructing our ISS-Net. As shown in Fig. 6, we conduct warm-up training under two configurations: one incorporating click loss and the other without it. To quantify the performance improvements

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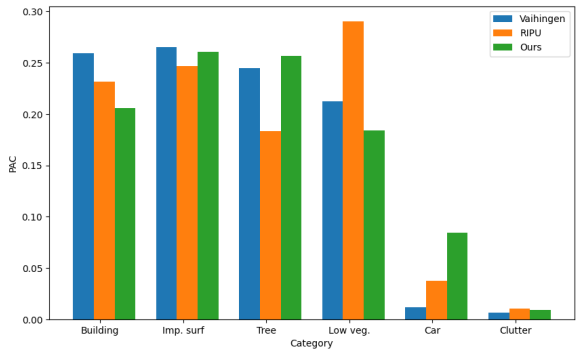



Fig. 11. Percentage of annotations per class (PAC) of the Vaihingen dataset, RIPU, and our method.

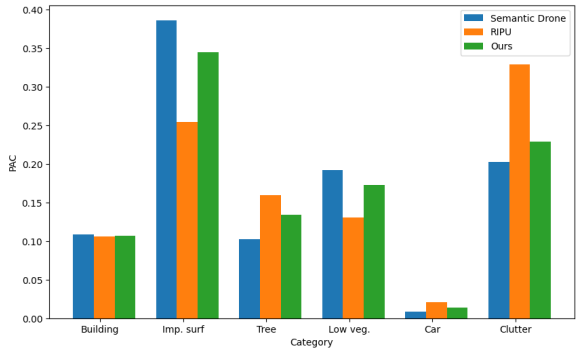



Fig. 12. PAC of the Semantic Drone dataset, RIPU, and our method.

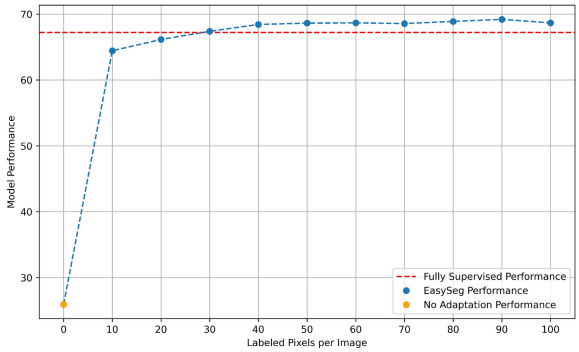



Fig. 13. Tradeoff between labeling cost and model performance for the Potsdam → Vaihingen task.

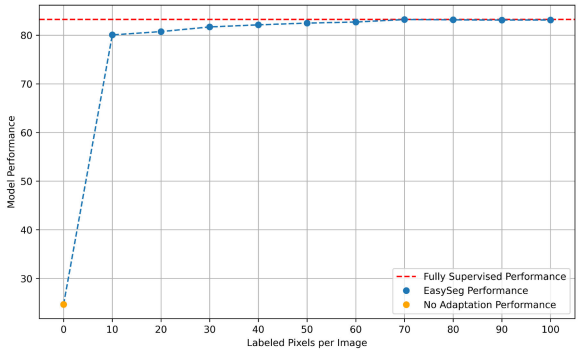



Fig. 14. Tradeoff between labeling cost and model performance for the Potsdam → Semantic Drone task.

of interactive predictions $p^s_{\text{iseg}}$ in comparison to automatic predictions $p^s$, we calculate the gain using the following formula:

$$\text{gain} = \frac{\text{mIoU}\left(p^s_{\text{iseg}}\right) - \text{mIoU}(p^s)}{\text{mIoU}(p^s)} \tag{16}$$

where $\text{mIoU}(p)$ represents the mIoU of the prediction $p$.

Given that the number of sampled clicks during training is random, the gain varies across iterations, as shown in Fig. 6. The results show that when click loss is employed, the prediction accuracy improves based on the interactive input after the 1000th training iteration, indicating that the model has learned the capability of interactive refinement. Conversely, without click loss, the gain always approaches zero and may even turn negative, indicating that the refinement information from interactive input is either ignored or disruptive. In our experiments, we set the weight $\lambda_{\text{click}}$ of click loss to 2.0 to encourage the model to prioritize interactive information during training. In Fig. 6, it can be observed that after 1000 iterations, a high-weighted click loss becomes nearly zero, suggesting that the weight $\lambda_{\text{click}}$ is reasonable and does not require any particular design.

## V. Discussion

We illustrate our $F1$ score improvements in each class compared to the baseline RIPU in Figs. 9 and 10. In Fig. 9, the proposed method demonstrates substantial improvements over the baseline RIPU in various categories for the Potsdam → Vaihingen task, especially in the clutter class, where our Deeplabv2-ResNet101 based (V2-based) model achieves a 30.34% mIoU improvement, and our Deeplabv3plus-ResNet101 based (V3-based) model achieves a 15.84% mIoU improvement. In Fig. 10, $F1$ score improvements are observed across all categories for the Potsdam → Semantic Drone task, especially in the building and clutter classes. For example, the V3-based model shows improvements of 4.45% and 2.47% in the building and clutter categories, respectively. We attribute these improvements to several factors: 1) the model's poor performance on the clutter class of the Potsdam dataset, where the appearance features significantly differ from those of the Vaihingen dataset, such as sandy soil in the Potsdam dataset and containers in the Vaihingen dataset. 2) Many objects in the Semantic Drone dataset cannot be learned from the Potsdam dataset due to object diversity, particularly the Building and Clutter classes. Without error awareness, these differences can lead to model misunderstandings and limited performance in these classes. However, our labeling strategy effectively corrects the model's fundamental misunderstandings about the objects, allowing it to progressively learn the target domain distribution and consequently increasing the value of acquired labels. Furthermore, our V2-based model shows more significant performance improvements in these classes than the V3-based one, suggesting that these misunderstandings are more severe in the weaker model. These experimental results indicate that significant but confident errors are a bottleneck for point-level methods, and our EasySeg effectively alleviates this problem.

The percentages of annotations per class (PACs) of the target dataset, RIPU, and our method are shown in Figs. 11 and 12 for the two tasks, respectively. First, both RIPU and

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our method efficiently allocate more annotations to the hard classes, which are rare or challenging classes, with limited labeling budgets. For instance, in the Vaihingen dataset, both RIPU and our method lead to a reduction in the PAC for the building category, while increasing the PAC for the car and clutter classes compared to the PAC of the Vaihingen dataset. Similarly, in the Semantic Drone dataset, both RIPU and our method result in decreased PAC for impervious surfaces, while reducing the PAC for the car and clutter relative to the PAC of the Semantic Drone dataset. Interestingly, RIPU exhibits higher PAC than our EasySeg in some categories within both datasets, yet fails to demonstrate superior performance in these categories. For example, in the Vaihingen dataset, RIPU achieves higher PAC than our EasySeg in the building, low vegetable, and clutter categories, but its performance in these three categories remains lower than that of EasySeg. This observation suggests that some annotations of RIPU have limited improvement in model performance, thus corroborating the high value of our obtained labels in the target domain. Based on these results, we argue that correcting obvious yet confident errors can mitigate biases in DA, and these passive labels acquired by interactive strategy also possess nonnegligible value compared to active labels acquired by query function.

We independently conducted ten experiments for each DA task, gradually increasing the labeled pixels per image from 10 to 100, following the settings described in Section IV-C. The tradeoff results between labeling cost and model performance of the two DA tasks are shown in Figs. 13 and 14. The experimental results demonstrate that the model's performance in the target domain improves gradually as the labeling costs increase. Notably, the most significant performance boost is observed with just ten labeled pixels per image. In the case of the Potsdam → Vaihingen task, our EasySeg model achieves near-convergence with 40 labeled pixels per image, exhibiting superior performance compared to fully supervised approaches. For the Potsdam → Semantic Drone task, the model nearly converges at 70 labeled pixels per image, gradually approaching the performance of the fully supervised model. This can be attributed to the significant domain gap between the Potsdam and Semantic Drone datasets, which limits the transfer of relevant knowledge from the source to the target domain.

## VI. Conclusion and Future Work

In this work, we introduce EasySeg, an error-aware ADA method for semantic segmentation in remote sensing imagery. Our framework emphasizes the efficient utilization of target domain data, aiming to address the challenge of error unawareness in point-level ADA methods. We propose the SFAL labeling strategy to make point-level ADA error aware while ensuring ease and efficiency in the labeling process. In addition, we design ISS-Net to generate accurate and dense pseudo-labels, harnessing the full potential of labeled target data. To fully leverage unlabeled target data, we apply consistency regularization to mitigate training noise and enhance model robustness. Experimental results demonstrate the superiority of our method, outperforming state-of-the-art methods with lower labeling costs, and even surpassing fully supervised models. Our experiments also yield an interesting finding: Error unawareness is one of the performance bottlenecks for point-level ADA methods.

In our EasySeg framework, there are still some aspects that exhibit potential for further exploration and investigation: 1) exploring the potential of boosting performance by integrating UDA methods into our warm-up training stage. 2) Exploring the potential of nonaverage point-level labeling strategies, especially in addressing class imbalance issues. 3) Exploring the potential of employing our EasySeg in other DA tasks, such as road extraction and change detection. In the future, we will delve deeper into these aspects to advance ADA for remote sensing imagery semantic segmentation.

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